Digital Systems and Microcontrollers

M26EC2.101, International Institute of Information Technology, Hyderabad

Question Bank 6

---

1. Mano: The IIITH Campus Rover

The rover Mano navigates eight locations, each represented by a 3-bit code:

Location	**Code**	**Location**	**Code**
Vindhya-A6	000	Bakul	001
OBH	010	Nilgiri	011
Main Gate	100	KRB	101
Kadamba	110	Himalaya	111

On each clock edge, the input bit determines the next location according to the transition table:

Current location	**Input 0**	**Input 1**
Vindhya-A6	Vindhya-A6	Bakul
Bakul	OBH	Nilgiri
OBH	Main Gate	Nilgiri
Nilgiri	Nilgiri	Vindhya-A6
Main Gate	Himalaya	KRB
KRB	Nilgiri	Kadamba
Kadamba	Himalaya	Kadamba
Himalaya	Bakul	Main Gate

(a) Draw the state diagram and construct the state table.

(b) Determine the final location for an all-ones input sequence and for the repeating sequence 0101.

(c) Derive minimized Boolean equations for the three next-state bits.

(d) Draw a gate-level implementation using D flip-flops and combinational logic. Assume synchronous reset places Mano at Vindhya-A6.

2. Parity Under a Forbidden Pattern

Construct a finite-state machine that accepts exactly those binary strings that contain an even number of 1s and do not contain the substring 101. The empty string is accepted.

(a) Construct a deterministic finite-state machine for this language.

(b) Derive next-state equations and draw a gate-level circuit with an accepting-state output.

Hint: Track parity and the suffix information needed to detect 101.

3. Histories That Must Remain Distinguishable

A binary string is accepted if every occurrence of 00 has at least one 1 somewhere after it before the string ends. Equivalently, reject a string if it contains 00 and the suffix after its last 00 contains no 1. The empty string and strings with no occurrence of 00 are accepted.

(a) Construct a deterministic finite-state machine for this language.

(b) Implement the machine as a synchronous circuit.

Hint: A state must preserve whether a recent 00 still needs to be followed by a 1.

4. Product Automaton with Three Constraints

Design a machine that accepts a binary string if and only if all of the following conditions hold:

(i) It contains an odd number of 0s.

(ii) Its final two bits are different.

(iii) It does not contain 111.

The empty string is rejected.

(a) Construct a product-state machine or an equivalent direct construction.

(b) Remove unreachable states.

(c) Give the resulting state table and identify the accepting states.

(d) Derive Boolean next-state and output equations for your chosen state encoding.

(e) Draw a gate-level circuit implementation.

Hint: Track zero parity, the previous bit, and the run length of trailing 1s.

5. Overlapping Pattern Detector: Mealy vs. Moore

Design both a Mealy machine and a Moore machine that detect the bit pattern 1011. Overlapping occurrences must be detected.

(a) Draw both state diagrams.

(b) Compare the number of states and output timing.

(c) Derive next-state and output equations.

(d) Draw both circuits.

(e) Trace the input sequence 1011011 cycle by cycle and clearly state when each output is asserted.

Hint: Each state can represent the longest suffix of the bits seen so far that is also a prefix of 1011.

6. Synthesize a Security Protocol

A binary string is accepted if and only if it contains 1101 and does not contain 001 anywhere in the string. The empty string is rejected. Once 001 has appeared, the string must remain rejected regardless of later input.

(a) Construct a deterministic finite-state machine.

(b) Minimize the machine.

(c) Derive next-state equations and acceptance logic.

(d) Draw the circuit.

(e) Test the design on the strings 1101, 0011101, 1101100, 11001101, and 11101.

Hint: Combine a detector for 1101 with a permanent reject condition triggered by 001; account

for overlapping prefixes.

7. Mealy Finite-State Machine: From Table to Circuit

The following is a five-state Mealy finite-state machine. Each table entry is written as next state / output. Assume the reset state is A.

State	$x = 0$	$x = 1$
A	B / 0	C / 0
B	B / 0	D / 1
C	E / 0	C / 0
D	B / 1	E / 0
E	A / 0	D / 1

(a) Draw the state diagram corresponding to the given state table. Label each transition with input/output.

(b) Assign distinct 3-bit binary codes to states A through E. Using your chosen encoding, derive Boolean equations for each next-state bit and for the output y as functions of the present-state bits and input x. Clearly state your encoding.

(c) Draw the gate-level circuit using D flip-flops and combinational logic. Show how reset places the machine in state A.

8. Detector for Two 0s and Two 1s: Mealy vs. Moore

Design both a Mealy machine and a Moore machine that produce output 1 whenever the most recent four input bits contain exactly two 0s and two 1s, in any order. The output is 0 until at least four input bits have been received. Overlapping occurrences must be detected.

(a) Draw the state diagram for the Mealy machine and give its state table.

(b) Draw the state diagram for the Moore machine and give its state table.

(c) Compare the number of states and the output timing of the two implementations.

(d) Derive next-state and output equations, then draw a gate-level circuit for each machine.

(e) Trace the input sequence 0011001101 cycle by cycle and show the output of each implementation.

Hint: The decision depends on the composition of the most recent four bits, not on matching one fixed pattern. Retain enough recent-input information to evaluate each shifted window.